Bias in the Machine: Investigating Fairness in AI-Hiring Based Systems

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1) Intro + Background

AI has quickly become a part of how companies screen job candidates, especially in the early stages of hiring. This process has become extremely widely adopted. At first glance, using AI might seem like a way to make hiring more efficient and objective, but as these tools have become more common, so have concerns about how fair they really are. A lot of the time, these AI-based hiring systems are trained on old data that reflect human biases, which means they can end up reinforcing and reflecting those same biases instead of removing them (Vivek, 2023; Hussien et al., 2024).

Researchers and critics have shown that AI-based hiring systems often behave in ways that reflect broader inequalities or biases in society. Even though companies might advertise these new tools as neutral or data-driven, studies show they can give some applicants an unfair advantage (or disadvantage), especially based on race, gender, or other defining traits (Drage & Mackereth, 2022; Albaroudi et al., 2024). And even when companies try to fix these issues, it's not always clear what “fairness” actually means in context. One company's version of a fair AI-based algorithm may still result in the perpetuation of past biases. Others may decide to rely on a measure of fairness that, while statistically valid, doesn't take into account qualitative indicators, actual experiences of discrimination or systemic barriers faced by marginalized groups.

This gap between technical, or defined fairness and how fairness is experienced in actual experience is one of the most common challenges discussed in the current debates about AI in hiring (Kassir et al., 2023). While data scientists and the developers of these tools often think about fairness using metrics like statistical parity or balanced error rates, people who are impacted by these decisions might focus more on how the process feels based on their experience. These different ways of defining fairness for something like an AI-based hiring system creates misunderstandings and tension between how these systems are designed and how they are received in the real world, and what the perception of applying to jobs or going for an interview becomes as a result of that.

What's missing in this world of AI-based hiring systems is a clearer picture of how these systems actually work in the real world, and how the people affected by them such as job seekers understand what fairness looks like (Kassir et al., 2023; Chen, 2023). Current studies focus a lot on one side of the issue either just the algorithm's outcomes or just public perception, but rarely both. The aim of my research is to close those gaps by collecting both quantitative data and experience-based personal insights to get a better sense of what's really going on behind the AI-based hiring systems in general. This will allow for a better understanding on not only where these systems need to improve based on their faults, but also how to move forward with designing A-based hiring tools that are not only effective but also fair.

2) Research Questions and Hypotheses

Main Research Question: Do AI hiring systems replicate or amplify bias in employment decision-making, and what ethical challenges arise from their use?

Sub-questions:

- What forms of bias, if any, are embedded in algorithmic hiring tools?
- How do key stakeholders (developers, employers, applicants) understand and address algorithmic fairness?
- How do outcomes differ across platforms, industries, and demographic contexts?

Hypotheses:

- **H1:** AI hiring systems will show noticeable bias in callback rates based on demographic indicators like gender or race.
- **H2:** Different groups involved in the hiring process (like developers, employers, and applicants) will have different views of what fairness means, depending on their role and perspective.
- **H0 (Null):** AI hiring systems will not show any meaningful differences in how they treat applicants from different demographic backgrounds.

3) Research Methodology

This study uses a mixed-methods design, containing methods such as an experimental audit study, algorithm output analysis, and semi-structured interviews. The mixed-methods approach gives the ability to examine not just the observable hiring outcomes, but also the perceptions behind them. The audit studies and algorithm output analysis will reveal measurable, quantitative patterns of bias, while interviews will provide context for how fairness and discrimination are understood and experienced by people who are impacted in one way or another in the hiring process. This is to understand both the “what” and the “why”

This design also allows an approach that is able to handle the complexity or real-world situations, as the mixed-methods approach allows for a lot of flexibility . The ability to mix numerical data with experiences based response allows for a better understanding not only where and how biases occur but also why they manage to persist and how they are rationalized. Another advantage of using both approaches is that it makes the findings of this research study more applicable and interesting to multiple audiences, such as academics, HR professionals, and even job seekers. It gives people looking for data and those who just want to understand the human stories behind those numbers what they are looking for at the same time. The goal of the research methods aren't just to point out what’s wrong, but also to build a picture of how people think about and try to fix the root of the problem. That’s something that is not possible through just numbers alone.

3.1 Quantitative Component:

- a) Audit Study: I will design an experiment involving the submission of 300–500 synthetic resumes to job postings that use AI-based screening methods. Each resume pair will be identical except for one altered demographic variable (a name suggesting gender or ethnicity for example). Callback rates will be recorded and statistically analyzed to find any connections.

- b) Algorithm Output Analysis: Along with the audit study I will analyze the outputs of AI hiring tools (using public datasets or simulated tools) to assess how different profiles are ranked. Regression models will be used to find out whether distinct features or characteristics predict hiring recommendations or outcomes based on the analysis.

3.2 Qualitative Component:

While the quantitative portion of this study helps to uncover patterns in AI behavior with numerical data, it is the qualitative component that captures the human experiences behind those patterns. Interviews allow us to hear directly from those designing, using, and being evaluated by these systems. That variety allows this part of the research to provide a far more personal look into what fairness means to different people, and how their perceptions of fairness or unfairness can shape trust in the hiring process, especially with AI-based tools in the picture for the foreseeable future.

- Semi-structured Interviews: Interviews will be conducted with 3 groups: AI developers, HR professionals, and job applicants. These interviews will explore how fairness is defined and implemented in each of their specific cases and roles, how stakeholders perceive bias, and what strategies are in place to mitigate. Responses will be coded thematically and then analyzed.

3.3 Sampling:

For the audit study, I will collect job ads from public platforms like LinkedIn and Indeed, focusing on postings or companies that are known for using AI-based tools. The postings will span across a variety of industries and job levels to increase the diversity in context and the diversity of the data. This will help test whether certain biases appear more strongly in some sectors than others, or for people applying for jobs at certain levels etc.

For interviews, I will use platforms such as professional networks or job-seeker forums to reach out to potential participants. I am aiming for about 20 interviewees, spread across the three main categories: developers, HR professionals, and job applicants. This structure will give a balanced view of how these hiring tools are understood and experienced from different angles by different people in different positions. I will also try to ensure demographic diversity across

the participants so that multiple perspectives are represented, especially from groups that are at a higher risk of being included in the bias. I will choose participants based on how relevant they are to the topic and whether they are open to having an in-depth conversation.

3.4 Integration Plan:

The quantitative data in this study will show whether bias exists and what kinds of patterns show up across different groups and situations. At the same time, the qualitative data will help explain how people understand and respond to those patterns. By combining both these types of information, this study can explore not just whether there is bias in the first place, but also how it is noticed or perceived, or even overlooked by the people involved. Looking at the numbers alongside personal experiences helps make the findings more meaningful. It also helps show how different parts of the hiring process connect, from automated decisions to human judgment.

3.5 Ethical Considerations:

Since this project includes both fake resume testing and interviews with real people, I plan to be very careful about ethics. Firstly, resumes will not be used to mislead employers or try to get actual jobs. They will only be sent to listings that clearly use AI to screen applications, and I will pull them out right away if they get selected.

For the interviews, I will make sure everyone knows what the research is about and i”ll make sure that they’re joining voluntarily and that they are aware of this. I will keep all personal information private and store everything securely. Everyone will get a consent form and can choose to stop at any time. I will also apply for Institutional Review Board (IRB) approval to make sure the entire project follows university guidelines, as in my research I have found that it is a standard and expected practice for any research involving human participants.

3.6 Timeline:

- Month 1: I will finalize the research design and develop the things that I will use for collecting data, such as drafting the synthetic resumes for the audit study, writing the interview questions, and submitting for IRB approval. I will also test some of the resume formatting and data collection sheets to make sure everything is usable.
- Months 2–3: At this stage I will start sending out the synthetic resumes and tracking responses for the audit study. I will also start to reach out to potential interview participants and start conducting interviews. I think during this time the first half of the interviews can be concluded.
- Month 4: During this month the second half of the interviews will be concluded and I will shift focus to analyzing the data collected so far. This would include things like reviewing and coding the interview transcripts thematically and starting to analyze the resume data based on callbacks.
- Month 5: This is when I will do most of the analysis. I will finish all the regression analysis for the audit study and complete the sorting and coding of the interviews. Then I would start looking at the data together to see if there is any possible overlap.
- Month 6: After all the research work is done, this last period of time will be dedicated to writing the report based on the research. This would include all of the standard report writing steps such writing and revising drafts and creating charts or visuals for the final product.

4) Significance and Impact

This research helps move the conversation around AI-based hiring systems and algorithmic hiring forward by blending the understanding brought by both numbers and stories. Using audit studies and algorithm output analysis gives a quantitative, data-driven view of how bias shows up in hiring decisions. The interviews are just as important because they help explain what these systems feel like in the real-world setting for the people building them, using them, and being affected by their decision.

Beyond just spotting bias, this research aims to understand how people think about the concept of fairness and what kinds of solutions they've tried in challenging situations. That kind

of insight really matters, especially as more companies use AI in hiring but do not always have the tools to “measure” fairness issues that come with it. The results could help inform better hiring practices all over the world, create cause for more thoughtful design of AI tools, and even clear up the communication between all the people involved with these tools.

This research also fits into a much broader conversation about how AI should be used in society as a whole. It is related to debates around transparency and accountability, and also whether these automated systems are making things better or worse. Instead of staying at the level of critique that is already there in society, this research tries to offer real suggestions that people in different roles can actually use in regard to AI-based hiring systems.

Potential Impacts:

- Policy & Ethics: Offer guidelines for ethical AI hiring design and deployment.
- Industry Practices: Provide insights for HR teams on where bias enters and how to audit for it.
- Public Awareness: Give applicants the knowledge to better understand how they are evaluated in today's day and age.

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